

Proofs and Properties of Equality and Congruence (Part III)

Prove the following statements with two-column proofs.

① If $\angle A$ and $\angle B$ are both right angles, then they are congruent.

Statement	Reason
I) $\angle A$ & $\angle B$ are rt. $\angle$'s	Given.
II) $m\angle A = 90^\circ$ & $m\angle B = 90^\circ$	Def. of rt. $\angle$'s.
III) $m\angle A = m\angle B$	Trans. Prop of =.
IV) $\angle A \cong \angle B$	Def. of $\cong \angle$'s.

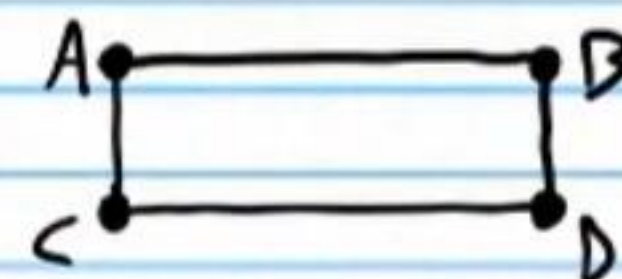
Right Angle Congruence Theorem: If two angles are both right angles, then they are congruent.

② If $\angle A$ and $\angle B$ are both straight angles, then they are congruent.

Statement	Reason
I) $\angle A$ & $\angle B$ are straight $\angle$'s.	Given.
II) $m\angle A = 180^\circ$ & $m\angle B = 180^\circ$	Def. of straight $\angle$'s.
III) $m\angle A = m\angle B$	Trans. Prop of =.
IV) $\angle A \cong \angle B$	Def. of $\cong \angle$'s.

Perpendicular Lines: Lines that intersect to form right angles.

③ If $\overline{AB}$ and $\overline{BD}$ are perpendicular, and $\overline{AC}$ and $\overline{CD}$ are perpendicular, then $\angle ABD$ and $\angle ACD$ are congruent.



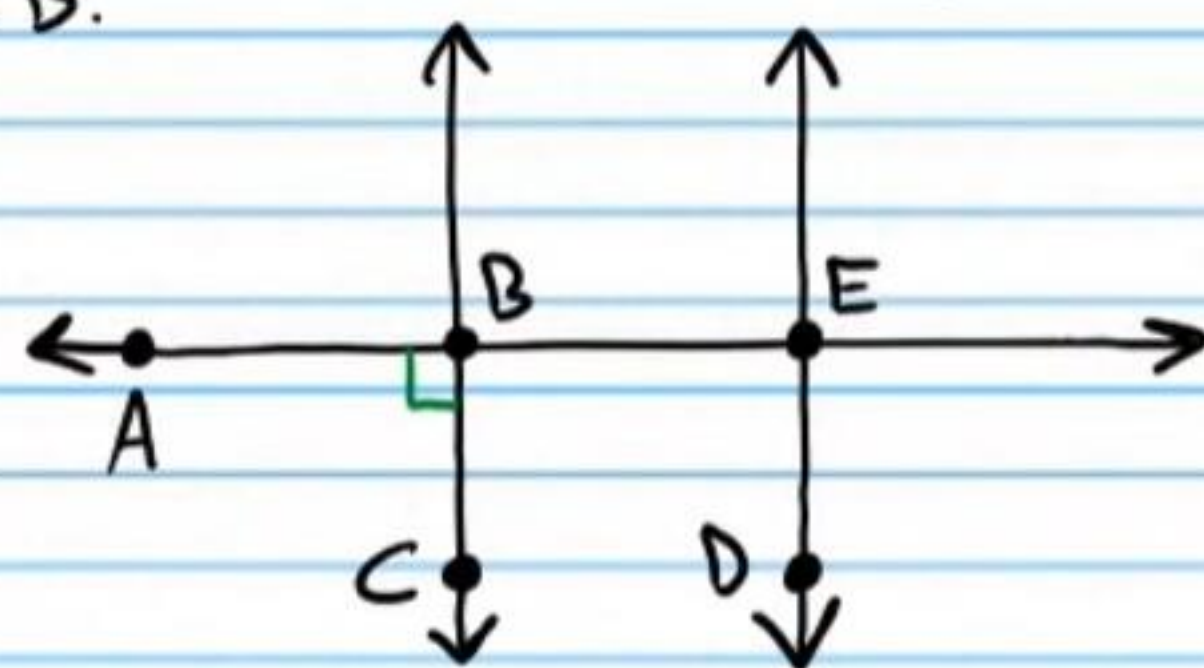
Proof I (without the Right $\angle \cong$ Thm)

Statement	Reason
I) $\overline{AB} \perp \overline{BD}$ & $\overline{AC} \perp \overline{CD}$.	Given.
II) $\angle ABD$ & $\angle ACD$ are rt $\angle$'s.	Def of $\perp$.
III) $m\angle ABD = 90^\circ$ & $m\angle ACD = 90^\circ$	Def. of rt. $\angle$'s
IV) $m\angle ABD = m\angle ACD$	Trans. Prop of =.
V) $\angle ABD \cong \angle ACD$	Def. of $\cong \angle$'s.

Proof II (with the Right $\angle \cong$ Thm)

Statement	Reason
I) $\overline{AB} \perp \overline{BD}$ & $\overline{AC} \perp \overline{CD}$.	Given.
II) $\angle ABD$ & $\angle ACD$ are rt $\angle$'s.	Def of $\perp$.
III) $\angle ABD \cong \angle ACD$	Right $\angle \cong$ Thm.

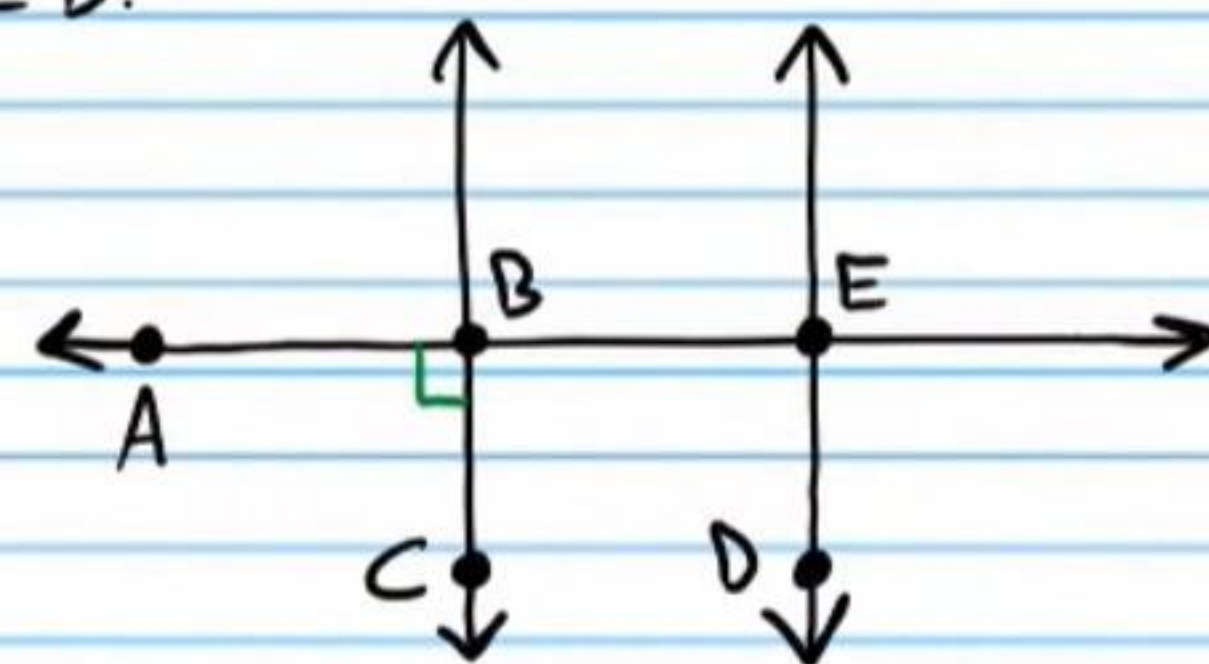
④ If $\overleftrightarrow{BE} \perp \overleftrightarrow{DE}$ and $\angle ABC$ is a right angle, then $\angle ABC \cong \angle DEB$.



Proof I (without the Right $\angle \cong$ Thm).

Statement	Reason
I) $\overleftrightarrow{BE} \perp \overleftrightarrow{DE}$	Given.
II) $\angle DEB$ is a rt. $\angle$.	Def of $\perp$.
III) $\angle ABC$ is a rt. $\angle$.	Given.
IV) $m\angle DEB = 90^\circ$ & $m\angle ABC = 90^\circ$.	Def of rt. $\angle$'s.
V) $m\angle ABC = m\angle DEB$	Trans. Prop of $=$.
VI) $\angle ABC \cong \angle DEB$	Def. of $\cong \angle$'s.

If $\overleftrightarrow{BE} \perp \overleftrightarrow{DE}$ and $\angle ABC$ is a right angle, then $\angle ABC \cong \angle DEB$.



Proof II (with the Right $\angle \cong$ Thm)

Statement	Reason
I) $\overleftrightarrow{BE} \perp \overleftrightarrow{DE}$	Given.
II) $\angle DEB$ is a rt. $\angle$.	Def of $\perp$.
III) $\angle ABC$ is a rt. $\angle$.	Given.
IV) $\angle ABC \cong \angle DEB$	Right $\angle \cong$ Thm.

⑤ If $\angle A$ and $\angle B$ are both supplementary to $\angle C$, then $\angle A \cong \angle B$.

Statement	Reason
I) $\angle A$ is supplementary to $\angle C$ and $\angle B$ is supplementary to $\angle C$.	Given.
II) $m\angle A + m\angle C = 180^\circ$ and $m\angle B + m\angle C = 180^\circ$.	Def. of supp $\angle$'s.
III) $m\angle A + m\angle C = m\angle B + m\angle C$	Trans. Prop of =.
IV) $m\angle A = m\angle B$	Subt. Prop of =.
V) $\angle A \cong \angle B$	Def. of $\cong \angle$'s.

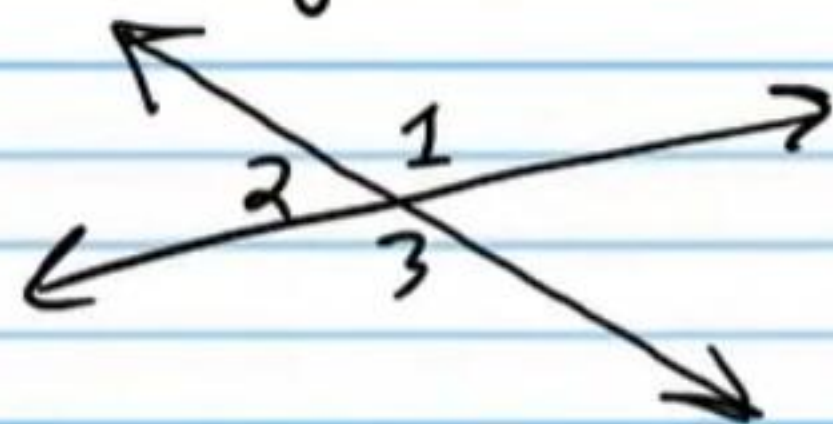
Congruent Supplements Theorem: If two angles are supplementary to the same angle, then they are congruent.

⑥ If $\angle A$ and $\angle B$ are both complementary to $\angle C$, then $\angle A \cong \angle B$.

Statement	Reason
I) $\angle A$ is complementary to $\angle C$ and $\angle B$ is complementary to $\angle C$.	Given.
II) $m\angle A + m\angle C = 90^\circ$ and $m\angle B + m\angle C = 90^\circ$	Def. of comp $\angle$'s.
III) $m\angle A + m\angle C = m\angle B + m\angle C$	Trans. Prop of =.
IV) $m\angle A = m\angle B$	Subt. Prop of =.
V) $\angle A \cong \angle B$	Def. of $\cong \angle$'s.

Congruent Complements Theorem: If two angles are complementary to the same angle, then they are congruent.

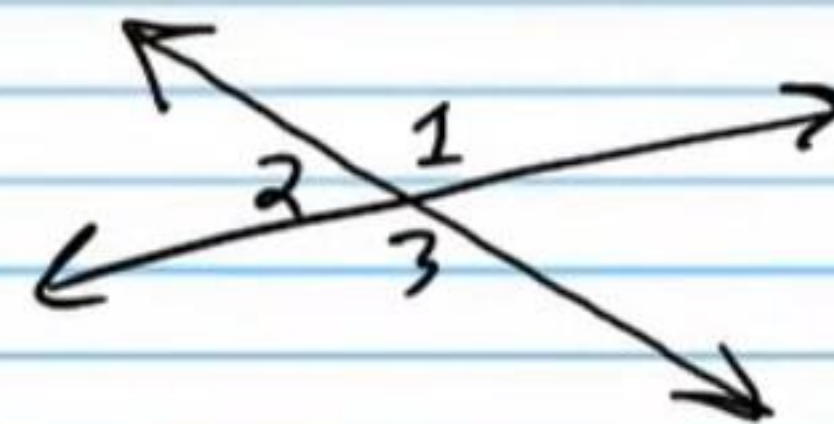
⑦ Vertical angles are congruent.



Proof I (without the $\cong$ Supplements Thm)

Statement	Reason
I) $\angle 1$ & $\angle 3$ are vert. $\angle$'s	Given (diagram).
II) $\angle 1$ & $\angle 2$ form a linear pair and $\angle 2$ & $\angle 3$ form a linear pair.	Given (diagram)
III) $\angle 1$ & $\angle 2$ are supplementary $\angle$'s and $\angle 2$ & $\angle 3$ are supplementary $\angle$'s.	Linear Pair Post.
IV) $m\angle 1 + m\angle 2 = 180^\circ$ and $m\angle 2 + m\angle 3 = 180^\circ$.	Def. of Supp $\angle$'s.
V) $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$	Trans. Prop of $=$.
VI) $m\angle 1 = m\angle 3$	Subt. Prop of $=$.
VII) $\angle 1 \cong \angle 3$	Def of $\cong \angle$'s

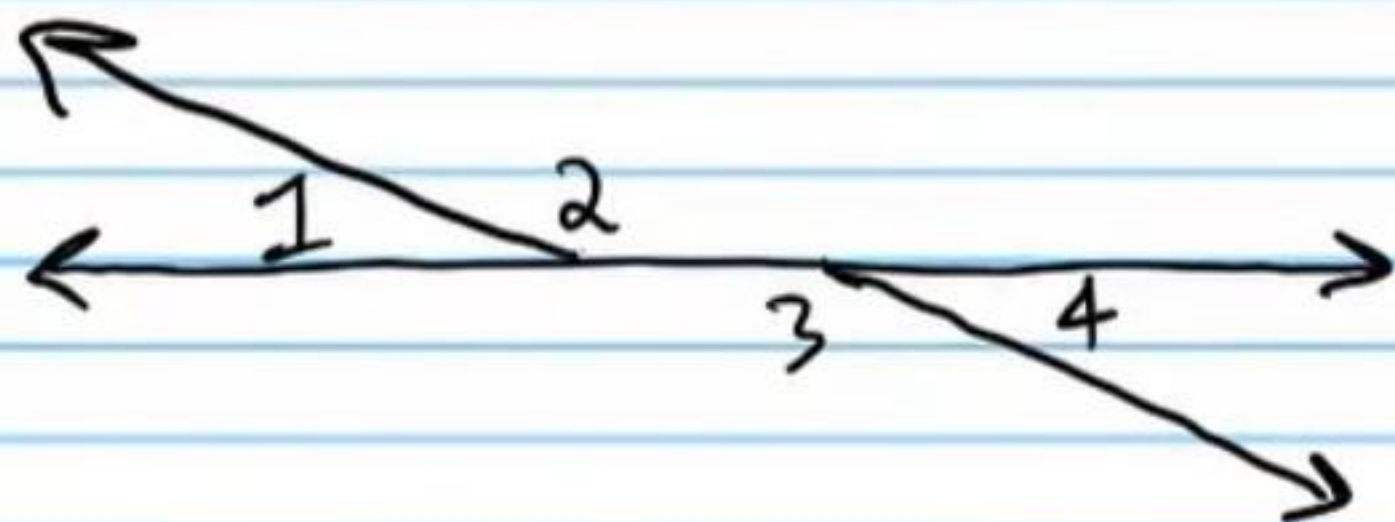
Vertical angles are congruent.



Proof II (with the $\cong$ Supplements Thm)

Statement	Reason
I) $\angle 1$ & $\angle 3$ are vert. $\angle$'s	Given (diagram).
II) $\angle 1$ & $\angle 2$ form a linear pair and $\angle 2$ & $\angle 3$ form a linear pair.	Given (diagram)
III) $\angle 1$ & $\angle 2$ are supplementary $\angle$'s and $\angle 2$ & $\angle 3$ are supplementary $\angle$'s.	Linear Pair Post.
IV) $\angle 1 \cong \angle 3$	$\cong$ Supp Thm.

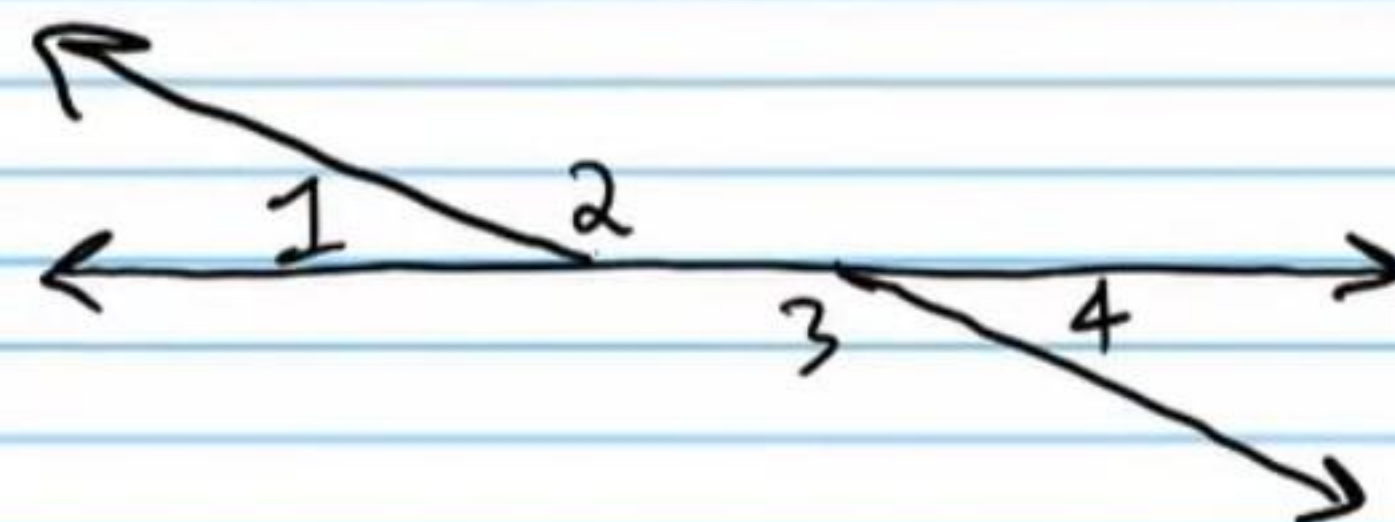
⑧ If $\angle 2$ and $\angle 4$ below are supplementary, then $m\angle 2 = m\angle 3$.



Proof I (without the $\cong$ Supplements Thm)

Statement	Reason
I) $\angle 3$ & $\angle 4$ form a linear pair.	Given (diagram).
II) $\angle 3$ & $\angle 4$ are supplementary $\angle$'s	Linear Pair Post.
III) $m\angle 3 + m\angle 4 = 180^\circ$	Def. of supp $\angle$'s
IV) $\angle 2$ & $\angle 4$ are supplementary $\angle$'s	Given.
V) $m\angle 2 + m\angle 4 = 180^\circ$	Def. of supp $\angle$'s
VI) $m\angle 3 + m\angle 4 = m\angle 2 + m\angle 4$	Trans. Prop of =.
VII) $m\angle 3 = m\angle 2$	Subt. Prop of =.

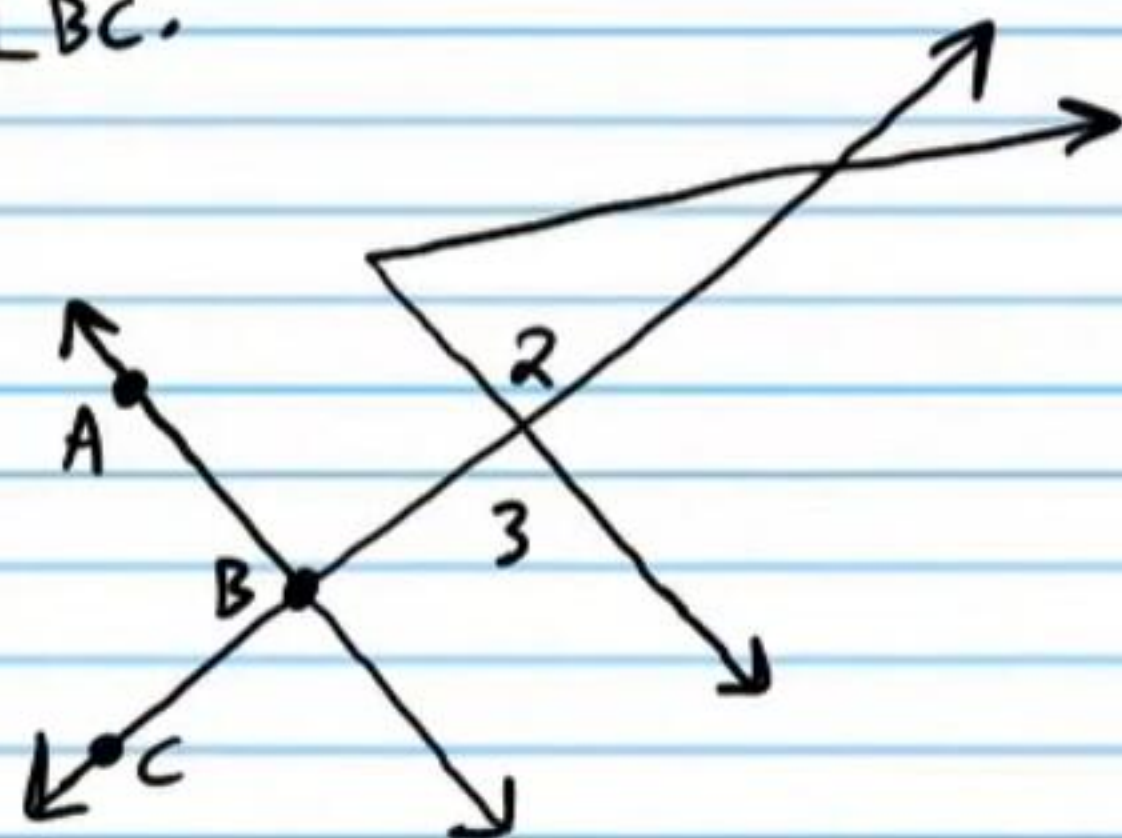
If $\angle 2$ and $\angle 4$ below are supplementary, then $m\angle 2 = m\angle 3$.



Proof II (with the $\cong$ Supplements Thm)

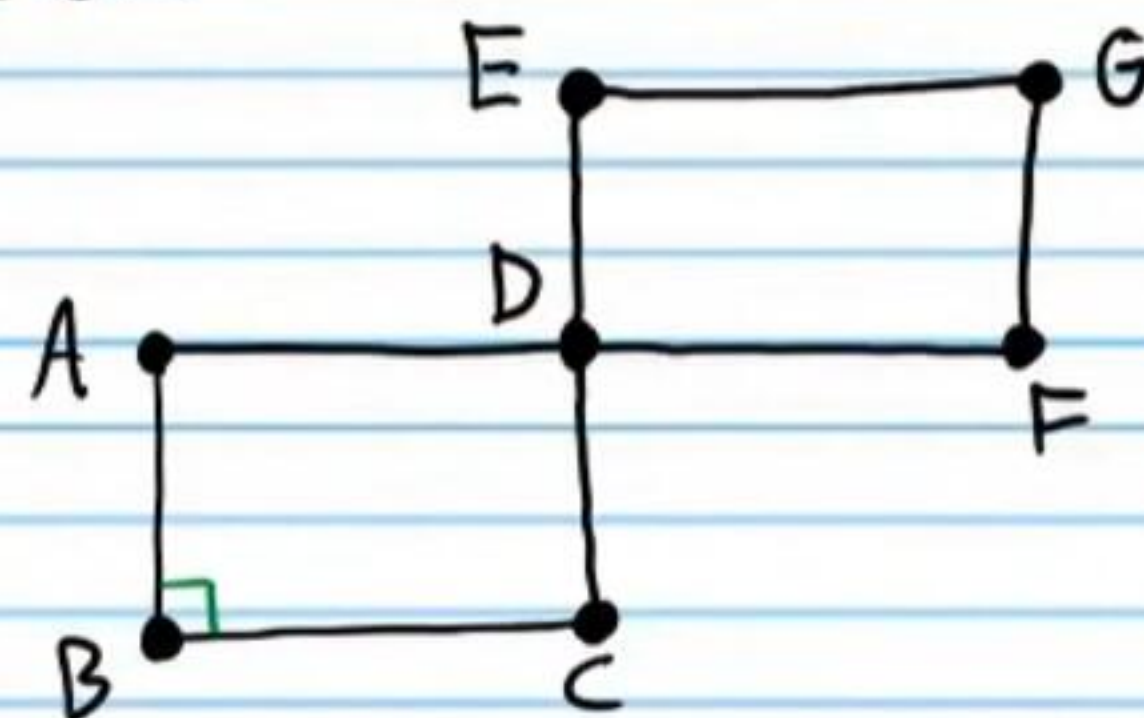
Statement	Reason
I) $\angle 3$ & $\angle 4$ form a linear pair.	Given (diagram).
II) $\angle 3$ & $\angle 4$ are supplementary $\angle$'s	Linear Pair Post.
III) $\angle 2$ & $\angle 4$ are supplementary $\angle$'s	Given.
IV) $\angle 2 \cong \angle 3$	$\cong$ Supp. Thm.
V) $m\angle 2 = m\angle 3$	Def. of $\cong \angle$'s.

⑨ If $m\angle ABC = m\angle 2$ and $\angle 3$ is a right angle, then $\overleftrightarrow{AB} \perp \overleftrightarrow{BC}$.



Statement	Reason
I) $\angle 3$ is a rt. $\angle$.	Given.
II) $m\angle 3 = 90^\circ$.	Def. of rt. $\angle$.
III) $m\angle 3 = m\angle 2$	Vert. $\angle$'s Thm.
IV) $m\angle 2 = 90^\circ$	Trans. Prop of =.
V) $m\angle ABC = m\angle 2$	Given.
VI) $m\angle ABC = 90^\circ$	Trans. Prop of =.
VII) $\angle ABC$ is a rt $\angle$.	Def. of rt $\angle$.
VIII) $\overleftrightarrow{AB} \perp \overleftrightarrow{BC}$	Def of $\perp$ lines.

⑩ If $m\angle ABC = m\angle ADC$ below and $m\angle EDF = m\angle EGF$, then $\overleftrightarrow{EG} \perp \overleftrightarrow{GF}$.



Statement	Reason
I) $\angle ABC$ is a rt $\angle$.	Given (diagram).
II) $m\angle ABC = 90^\circ$.	Def. of rt $\angle$.
III) $m\angle ABC = m\angle ADC$	Given.
IV) $90^\circ = m\angle ADC$	Trans. Prop of =.
V) $m\angle ADC = m\angle EDF$	Vert. $\angle$'s Thm.
VI) $m\angle EDF = 90^\circ$	Trans. Prop of =.
VII) $m\angle EDF = m\angle EGF$	Given.
VIII) $m\angle EGF = 90^\circ$	Trans. Prop of =.
IX) $\angle EGF$ is a rt $\angle$.	Def of rt $\angle$.
X) $\overleftrightarrow{EG} \perp \overleftrightarrow{GF}$	Def of $\perp$.